WORKED SOLUTION

STEP III 2016, Question 6

Pure • Hyperbolic Functions

STEP III 2016, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q6

The key idea

Multiplying $\operatorname{sech} x = a\tanh x + b$ through by $\cosh x$ gives $a\sinh x + b\cosh x = 1$, which is exactly the combination in the opening part. So the first job is the hyperbolic version of the familiar "$R\cos(x+\alpha)$" trick. Unlike the trigonometric case, the form you get depends on how B compares with $\pm A$, and each form tells you at once how many solutions there are.

Solution

The forms of $A\sinh x + B\cosh x$

Expanding,

$$R\cosh(x+\gamma) = R\sinh\gamma\,\sinh x + R\cosh\gamma\,\cosh x,$$

so we need $R\sinh\gamma = A$ and $R\cosh\gamma = B$. Since $\cosh^2\gamma - \sinh^2\gamma = 1$, this forces $R^2 = B^2 - A^2$ and $\tanh\gamma = A/B$. When $B > A > 0$ both of these can be met: take

$$R = \sqrt{B^2 - A^2} > 0, \qquad \gamma = \operatorname{artanh}\frac{A}{B},$$

which exists because $0 < A/B < 1$. Then $\cosh\gamma = 1/\sqrt{1 - A^2/B^2} = B/R$ and $\sinh\gamma = \tanh\gamma\cosh\gamma = A/R$, so $R\cosh\gamma = B$ and $R\sinh\gamma = A$, as required. Hence $A\sinh x + B\cosh x = R\cosh(x+\gamma)$.

The other cases, with $A > 0$:

- $-A < B < A$: matching $R\sinh(x+\gamma) = R\cosh\gamma\sinh x + R\sinh\gamma\cosh x$ instead gives

$$A\sinh x + B\cosh x = \sqrt{A^2 - B^2}\,\sinh(x+\gamma), \qquad \gamma = \operatorname{artanh}\frac{B}{A}.$$

- $B = A$: $A(\sinh x + \cosh x) = Ae^x$.

- $B = -A$: $A(\sinh x - \cosh x) = -Ae^{-x}$.
- $B < -A$: applying the first result to $-A\sinh x - B\cosh x$ (with $-B > A$ and $-A < 0$; the working above only needed $|A| < B$) gives

$$A\sinh x + B\cosh x = -\sqrt{B^2 - A^2}\,\cosh(x + \gamma), \qquad \gamma = \operatorname{artanh}\frac{A}{B}.$$

The intersection equation

Since $\cosh x > 0$, the curves meet where

$$\operatorname{sech} x = a\tanh x + b \iff 1 = a\sinh x + b\cosh x. \qquad (*)$$

Part (i)

With $b > a > 0$, write $R = \sqrt{b^2 - a^2}$; then $(*)$ becomes

$$\cosh\left(x + \operatorname{artanh}\tfrac{a}{b}\right) = \frac{1}{\sqrt{b^2 - a^2}}.$$

If the curves intersect, this has a solution, so the right-hand side is at least 1. The solutions of $\cosh u = c$ with $c \geq 1$ are $u = \pm\operatorname{arcosh} c$, so

$$x = \pm\operatorname{arcosh}\left(\frac{1}{\sqrt{b^2 - a^2}}\right) - \operatorname{artanh}\frac{a}{b},$$

which is the required form.

Part (ii)

With $a > b > 0$ we are in the sinh case: $(*)$ becomes $\sqrt{a^2 - b^2}\,\sinh\left(x + \operatorname{artanh}\frac{b}{a}\right) = 1$. Since sinh is a bijection from $\mathbb{R}$ to $\mathbb{R}$, there is always exactly one intersection, at

$$x = \operatorname{arsinh}\left(\frac{1}{\sqrt{a^2 - b^2}}\right) - \operatorname{artanh}\frac{b}{a}.$$

Part (iii)

Count the solutions of $(*)$ in each case of the opening part (with $A = a > 0$, $B = b$):

- $b < -a$: the left-hand side $-\sqrt{b^2 - a^2}\cosh(x + \gamma)$ is negative, so there is no solution.
- $b = -a$: $-ae^{-x} = 1$ has no solution.
- $-a < b < a$: exactly one solution, as in (ii); the argument there works for any b in this range.
- $b = a$: $ae^{x} = 1$ has the single solution $x = -\ln a$.
- $b > a$: $\cosh(x + \gamma) = 1/\sqrt{b^2 - a^2}$ has two solutions if $1/\sqrt{b^2 - a^2} > 1$, one if it equals 1, and none if it is less than 1.

Ask yourself Why can't the cases with $b \leq a$ ever give two points?

In those cases the left-hand side of $(*)$ is a strictly monotonic function of x (a multiple of $\sinh(x+\gamma)$ or of $e^{\pm x}$), or is always negative, so it takes the value 1 at most once. Only the cosh form, which goes down and then up again, can reach 1 twice.

So the curves meet at two distinct points if and only if $b > a$ and $b^2 - a^2 < 1$, that is,

$$a < b < \sqrt{a^2+1}.$$

Part (iv)

Since $\operatorname{sech} x - (a\tanh x + b) = \operatorname{sech} x\,(1 - a\sinh x - b\cosh x)$ and $\operatorname{sech} x \neq 0$, the curves touch exactly where $g(x) = a\sinh x + b\cosh x - 1$ has a repeated zero, i.e. $g(x) = g'(x) = 0$. In the sinh and exponential cases $g'(x)$ is a multiple of $\cosh(x+\gamma)$ or e^x, never zero, so there are no repeated zeros. In the case $b > a$, $g(x) = R\cosh(x+\gamma) - 1$ and $g'(x) = R\sinh(x+\gamma)$, which vanish together only when $x + \gamma = 0$ and $R = 1$. Hence the curves touch if and only if

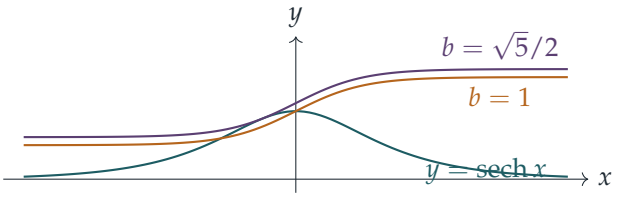
$$b > a \quad \text{and} \quad b^2 - a^2 = 1, \qquad \text{i.e.} \quad b = \sqrt{1+a^2}.$$

The point of contact has $x = -\gamma = -\operatorname{artanh}(a/b)$, so

$$y = \operatorname{sech}\left(\operatorname{artanh}\tfrac{a}{b}\right) = \sqrt{1 - \tfrac{a^2}{b^2}} = \frac{\sqrt{b^2-a^2}}{b} = \frac{1}{b} = \frac{1}{\sqrt{1+a^2}},$$

using $\operatorname{sech}^2 u = 1 - \tanh^2 u$ and $\operatorname{sech} u > 0$. As a check, the other curve gives $y = -a\cdot\frac{a}{b} + b = \frac{b^2-a^2}{b} = \frac{1}{b}$ too.

The picture below shows $a = \frac{1}{2}$: with $b = 1$ (amber) there are two intersections, and with $b = \sqrt{5}/2$ (plum) the curves touch at height $2/\sqrt{5}$.



Checks

Take $a = \frac{3}{4}$, $b = 1$, so that $b^2 - a^2 = \frac{7}{16} < 1$ and (iii) predicts two points. Then $\operatorname{arcosh}(4/\sqrt{7}) = \ln\big((4+3)/\sqrt{7}\big) = \ln\sqrt{7}$ and $\operatorname{artanh}\frac{3}{4} = \frac{1}{2}\ln 7 = \ln\sqrt{7}$, so the formula in (i) gives $x = 0$ and $x = -\ln 7 \approx -1.946$. At $x = 0$ both curves pass through $(0,1)$, as they should. A computer check of $(*)$ over many random a and b also confirmed the root counts in every case, and that the tangency condition gives a double root.

Common slips

- Forgetting the boundary cases $b = \pm a$, where neither cosh nor sinh form exists but the expression is a single exponential.
- In (iii), giving only $b^2 - a^2 < 1$ and leaving out $b > a$: for $b < a$ the curves still meet, but only once.
- Getting the sign of γ wrong: the solutions are $\pm\operatorname{arcosh}(\cdot)$ *minus* $\operatorname{artanh}(a/b)$.

Beyond STEP

The case split is governed by the sign of $B^2 - A^2$: the point (A, B) lies inside one of the regions bounded by the lines $B = \pm A$, or on those lines. This is the geometry of special relativity, where Lorentz boosts are hyperbolic "rotations" preserving $t^2 - x^2$ just as ordinary rotations preserve $x^2 + y^2$, and $B = \pm A$ plays the role of the light cone, the one boundary a boost can never move you across. For more hyperbolic functions and problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

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